📘 Chapter 2 – SMPS, BIOS, Booting & Troubleshooting

⚡ SMPS (Switch Mode Power Supply) – Revision

🔌 What is SMPS?

👉 SMPS provides **power to all components in a desktop computer**

💻 **Laptop vs Desktop:**

- Laptop Adapter → **~18.5V output**
- Desktop SMPS → Multiple voltages

🔋 SMPS Voltage Output (Important)

Wire Color	Voltage	Use
🔴 Red	+5V	USB, logic circuits
🟠 Orange	+3.3V	RAM, chipset
🟡 Yellow	+12V	CPU, GPU
🟣 Purple	+5V Standby	Always ON
🟢 Green	PS_ON	Power ON signal
⚫ Black	Ground	Return path

🔌 SMPS Connectors

1️⃣ **24-Pin ATX Connector**

👉 Supplies power to **motherboard**

2️⃣ **4-Pin CPU Connector**

👉 Supplies power to **processor (CPU)**

⭕ How Motherboard Starts (Power Flow)

🧩 Front Panel (F-Panel) Pins

These pins connect **cabinet buttons & LEDs** to motherboard.

🔗 4 Main Pairs:

- Power Button ⭕
- Restart Button 🔄
- Power LED 💡
- Hard Disk LED 💽

💡 Hard Disk Indicator (Important Concept)

👉 In laptops & desktops:

- HDD symbol = **Cylinder icon**
- If system hangs → HDD LED stops blinking

Example:

👉 If HDD light is stuck → System freeze or disk issue

🧠 BIOS (Basic Input Output System)

⭐ What is BIOS?

👉 BIOS is the **brain of the motherboard**

Function:

- Controls how motherboard works
- Starts system
- Checks hardware

🔄 Booting Process (Step-by-Step)

1. Power ON 🔌
2. BIOS starts 🧠
3. Company logo appears
4. Loading screen / cursor
5. BIOS checks storage device

💾 Booting Device

👉 Device from which OS loads:

- HDD / SSD / USB

📌 Important:

- OS is installed in **hard disk**
- BIOS decides **which device to boot from**

🧠 BIOS Stores:

- Date & Time ⏰
- Boot order
- BIOS password 🔐

🔧 Why BIOS is Used in Troubleshooting?

👉 Some issues are:

- Hardware level 🔩
- OS level 💻

👉 Hardware issues → solved using BIOS

⚙️ Chipset (Motherboard Controller)

⭐ What is Chipset?

👉 Controls communication between components

📜 Old vs New Design

Old:

- Northbridge → CPU, RAM, GPU
- Southbridge → USB, HDD

New:

👉 Single chipset (faster)

🔗 Chipset Controls:

- USB ports
- SATA ports
- PCIe slots
- Network (LAN)
- Audio
- CPU, RAM, GPU communication

🧠 Important:

👉 **BIOS + Chipset work together**

🔌 I/O Chip (Super I/O Controller)

What is I/O Chip?

👉 Chip with many pins (~128 pins)

Function:

- Transfers signals (like PS_ON)
- Controls input/output devices

⚡ Example Flow:

1. Power button pressed
2. PS_ON signal generated
3. I/O chip sends signal to BIOS

🔊 Audio Example (Important Logic)

👉 If audio not working:

1. System starts
2. I/O checks Power OK signal
3. If signal missing → No audio
4. Indicates **motherboard issue**

🔬 What is a Chip?

👉 Chip = **Integrated Circuit (IC)**

Function:

- Controls hardware
- Executes commands

🔗 **Control Flow:**

👉 Audio Chip ← Chipset ← BIOS ← I/O Chip

🔋 CMOS Battery

Voltage:

👉 ~3.3V

Function:

- Stores BIOS settings
- Keeps system time

🔥 **Important Concept:**

👉 Even when system OFF → BIOS settings saved

💾 **Boot Order**

Scenario:

👉 Two Hard Disks:

- HDD1 → OS installed
- HDD2 → Only data

👉 If BIOS selects HDD2:
❌ System will NOT boot

⚠️ **Error:**

👉 **“No Bootable Device Found”**

Causes:

- Wrong boot order
- HDD failure
- OS corrupt
- CMOS battery dead

🔋 **CMOS Battery Issue:**

👉 If BIOS resets again & again → battery dead

🔌 **USB Boot Issue**

👉 If USB is first boot device:

- System searches OS in USB
- If not found → error

🐢 System Slow Issue

Scenario:

- OS reinstalled
- Still slow

Possible Causes:

- BIOS not updated
- Chipset drivers missing
- Hardware limitation

🧠 Detailed Boot Process

1. Power ON
2. BIOS starts
3. BIOS checks HDD
4. Looks for:
    - Windows → **bootmgr**
    - Linux/Mac → **GRUB loader**

❌ **If file not found:**

👉 Error: **No Bootable Device**

🧠 **Concept:**

- CPU → Processing
- RAM → Temporary space

🖥️ NO DISPLAY Problem (VERY IMPORTANT)

📌 **Scenario:**

👉 Power ON but no display

🔍 **Step 1: Check Caps Lock LED**

👉 Keyboard Caps Lock LED is controlled by:
BIOS → Chipset → Keyboard

🧠 **Case Analysis:**

✅ **Case 1: Caps Lock LED ON/OFF Working**

👉 Meaning:

- BIOS OK
- Motherboard OK
- RAM OK

👉 Problem:

- Display issue
  - HDMI cable
  - Monitor

❌ **Case 2: Caps Lock NOT Working**

👉 Meaning:

- BIOS not working OR
- RAM faulty OR
- Motherboard issue

🔧 **Root Cause:**

- RAM failure
- BIOS failure

⚡ **BIOS Testing:**

👉 Check BIOS chip first pin → should get **3.3V**

- No voltage → BIOS faulty

🔌 NO POWER Problem (VERY IMPORTANT)

📌 **Scenario:**

👉 System not turning ON

🔍 **Step-by-Step Troubleshooting:**

1️⃣ **Check Power Supply**

- Plug & switch ON

2️⃣ **Desktop Check (SMPS Test)**

👉 Use **shorting method**:

- Connect Green + Black wire

👉 Result:

- Fan spins → SMPS OK
- No spin → SMPS faulty

3️⃣ **Laptop Check**

👉 Use multimeter:

- Output should be ~18.5V

4️⃣ **LED Behavior**

👉 Charger LED OFF after plug:

- Motherboard short circuit

🧠 NO POWER vs NO DISPLAY

Problem	Meaning
No Power	SMPS / Power issue
No Display	RAM / BIOS / Display issue

🎯 Interview Tip:

👉 If you explain both properly → interviewer knows:

- You understand **hardware deeply**

🚀 Real IT Support Mindset

✅ Always start with **power check**
✅ Observe LED indicators
✅ Think step-by-step
✅ Do not guess → verify
✅ Identify root cause fast

🏁 Final Understanding

👉 Boot Process:
Power → BIOS → Chipset → HDD → OS

👉 Problem Types:

- Power issue ⚡
- Display issue 🖥️
- Boot issue 💾

📘 Chapter 2 (SUMMARY)

⚡ SMPS (Power System)

👉 SMPS ka kaam:

- Computer ke sab components ko **proper voltage dena** ⚡
- AC → DC convert karna

🔋 **Important Voltages:**

- 🟡 12V → CPU, GPU
- 🔴 5V → USB
- 🟠 3.3V → RAM
- 🟣 5V Standby → Always ON

🔌 Power Flow (VERY IMPORTANT)

👉 System ON ka process:

1. Power button press ⭕
2. PS_ON signal (Green wire)
3. I/O chip signal send karta hai
4. BIOS start hota hai
5. System ON

🧠 BIOS (Brain of Computer)

👉 BIOS ka kaam:

- System start karna
- Hardware check karna
- Boot device select karna

👉 BIOS store karta hai:

- Date & Time ⏰
- Boot order 💾
- Password 🔐

🔄 **Booting Process**

👉 Step-by-step:

1. Power ON ⚡
2. BIOS start 🧠
3. Hardware check
4. HDD/SSD check
5. OS load

👉 Boot loader:

- Windows → bootmgr
- Linux → GRUB

⚙️ **Chipset**

👉 Chipset controls:

- CPU, RAM, GPU communication
- USB, SATA, PCIe

👉 Modern system:
👉 Single chipset (fast performance)

🔌 I/O Chip (Super I/O)

👉 Work:

- Power signal control
- Keyboard, fan, ports control

🔋 CMOS Battery

👉 Function:

- BIOS settings store karta hai
- Time maintain karta hai

👉 Problem:

- Battery dead → BIOS reset

💾 Boot Order Concept

👉 BIOS decides:

- Kaunsa device se OS load hoga

👉 Wrong boot order:
❌ No Bootable Device

🖥️ NO DISPLAY (Important)

✅ Case 1: Caps Lock Working

👉 Meaning:

- BIOS OK
- RAM OK
  👉 Problem: Display

❌ Case 2: Caps Lock NOT Working

👉 Meaning:

- RAM issue
- BIOS issue
- Motherboard issue

🔌 NO POWER (Important)

👉 Check:

1. SMPS
2. Adapter
3. Motherboard

👉 SMPS Test:
✔ Green + Black short
✔ Fan spin = OK

🧠 Final Logic

👉 Types of problems:

- Power Issue ⚡
- Display Issue 🖥️
- Boot Issue 💾

🎯 CONCLUSION

👉 Real IT Support Engineer ka mindset:

✅ Start with power check
✅ Observe LEDs
✅ Follow step-by-step process
✅ Identify root cause (guess nahi karna)

👉 Final line:
"Troubleshooting is not guessing, it is logical thinking + observation" 🔥

💼 🔥 Q & A

🔹 SMPS QUESTIONS

❓ Q1: What is SMPS?

👉 **Answer:**
SMPS (Switch Mode Power Supply) is a device that converts AC power into DC power and supplies different voltages to computer components like CPU, RAM, and motherboard.

❓ Q2: What are SMPS voltage outputs?

👉 **Answer:**

- 12V → CPU
- 5V → USB
- 3.3V → RAM
- 5V Standby → Always ON

❓ **Q3: What is PS_ON signal?**

👉 **Answer:**
PS_ON is a signal sent by motherboard to SMPS to start power supply.
It is controlled through **Green wire**.

❓ **Q4: How do you test SMPS?**

👉 **Answer:**

1. Disconnect SMPS
2. Short Green + Black wire
3. Check fan
   👉 Fan spins = SMPS working

🔹 **BIOS QUESTIONS**

❓ **Q5: What is BIOS?**

👉 **Answer:**
BIOS (Basic Input Output System) is firmware that initializes hardware and starts the booting process.

❓ **Q6: What does BIOS store?**

👉 **Answer:**

- Date & Time
- Boot order
- Password

❓ **Q7: Why BIOS is important?**

👉 **Answer:**
Because without BIOS, system cannot start or detect hardware.

🔹 **BOOTING QUESTIONS**

❓ **Q8: What is booting?**

👉 **Answer:**
Booting is the process of starting a computer and loading the operating system from storage into RAM.

❓ **Q9: What is boot device?**

👉 **Answer:**
Device from which OS loads (HDD, SSD, USB).

❓ **Q10: What is boot loader?**

👉 **Answer:**
A program that loads OS:

- Windows → bootmgr
- Linux → GRUB

🔹 **CMOS QUESTIONS**

❓ **Q11: What is CMOS battery?**

👉 **Answer:**
A small battery that stores BIOS settings and system time.

❓ **Q12: What happens if CMOS battery is dead?**

👉 **Answer:**

- BIOS resets
- Date/time incorrect
- Boot issues

🔹 **CHIPSET & I/O QUESTIONS**

❓ **Q13: What is chipset?**

👉 **Answer:**
Chipset manages communication between CPU, RAM, storage, and peripherals.

❓ **Q14: What is I/O chip?**

👉 **Answer:**
I/O chip controls input-output operations and power signals like PS_ON.

🔹 **TROUBLESHOOTING QUESTIONS (VERY IMPORTANT)**

🖥️ **NO DISPLAY**

❓ **Q15: System ON but no display, what will you do?**

👉 **Answer:**
Step-by-step:

1. Check monitor & cable
2. Check Caps Lock LED
3. If working → display issue

4. If not working → RAM/BIOS issue

🔌 NO POWER

❓ Q16: System not turning ON?

👉 **Answer:**

1. Check power supply
2. Test SMPS
3. Check motherboard
4. Check adapter (laptop)

💾 BOOT ISSUE

❓ Q17: No Bootable Device Found?

👉 **Answer:**
Possible causes:

- Wrong boot order
- HDD failure
- OS corrupt
- CMOS battery issue

❓ Q18: System slow after OS install?

👉 **Answer:**

- Install drivers
- Update BIOS
- Check hardware limitation

❓ **Q19: HDD LED not blinking?**

👉 **Answer:**

👉 System hang or disk issue

❓ **Q20: Charger LED OFF after plug?**

👉 **Answer:**

👉 Motherboard short circuit

❓ **Q21: Difference between No Power & No Display?**

👉 **Answer:**

No Power ⚡	**No Display** 🖥️
System not ON	System ON but no screen
SMPS issue	RAM/BIOS issue

❓ **Q22: How do you troubleshoot effectively?**

👉 **Answer:**

- Start with basics
- Follow steps
- Check indicators
- Verify results
